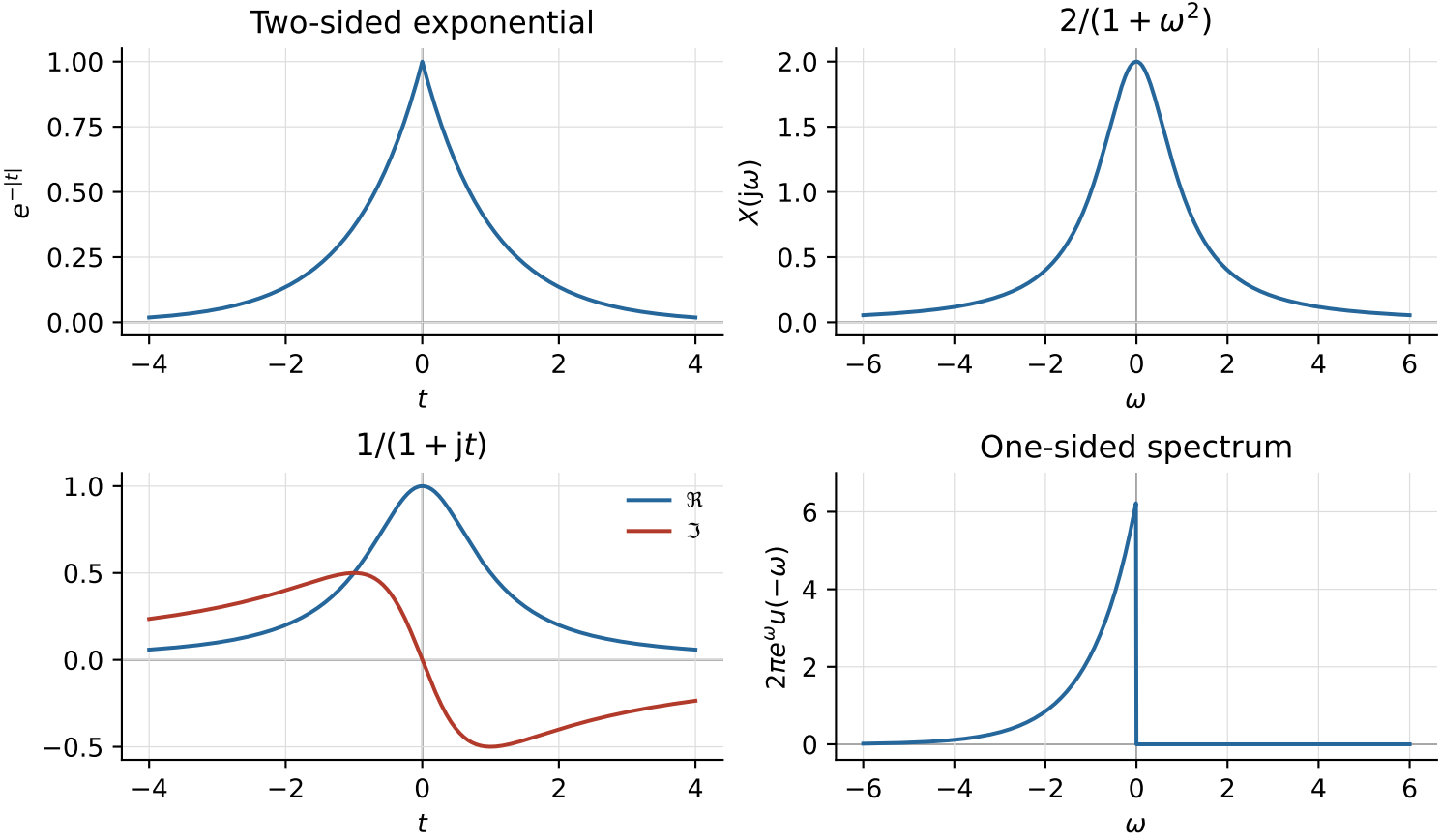
Two-sided exponential
$e^{-|t|}$
1.00
0.75
0.50
0.25
0.00
−4
−2
0
2
4
t
$2/(1+\omega^2)$
$X(j\omega)$
2.0
1.5
1.0
0.5
0.0
−6
−4
−2
0
2
4
6
ω
$1/(1+jt)$
$\Re$
$\Im$
1.0
0.5
0.0
−0.5
−4
−2
0
2
4
t
One-sided spectrum
$2\pi e^{\omega}u(-\omega)$
6
4
2
0
−6
−4
−2
0
2
4
6
ω